



CAVENDISH UNIVERSITY – ZAMBIA

ASSIGNMENT BRIEF AND FEEDBACK FORM

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ASSIGNMENT BRIEF

Project Title: Design and Development of Cavendish University Zambia Student Mental Health and Performance counselling web-based Application.

CHAPTER ONE: INTRODUCTION

1.0 Introduction

Mental health is an increasingly critical factor influencing students' academic performance and overall well-being. Universities worldwide are observing rising cases of stress, anxiety, depression, and other psychological disorders among students. Traditional counselling services, while effective in some cases, often face challenges such as limited accessibility, stigma, and insufficient staffing. Digital platforms offer an innovative approach to bridging this gap by providing timely, accessible, and confidential mental health support. A web-based counselling and performance support system can enable students to engage with professional support services while preserving anonymity and convenience, improving both academic outcomes and mental well-being.

1.1 Background

Globally, one in three universities students experience mental health challenges that affect their studies (WHO, 2022). Digital counselling solutions, such as online therapy platforms and mental health apps, have been effective in mitigating stress and anxiety in students. In Africa, while awareness of mental health is increasing, digital solutions remain underdeveloped. Zambian universities, including Cavendish University Zambia, face limitations in traditional counselling services due to low staffing, high student-to-counsellor ratios, and stigma surrounding mental health. This results in unmet mental health needs, negatively affecting students' academic performance, attendance, and social engagement. The development of a localized web-based counselling application aims to address these gaps, providing a solution tailored to students' needs and cultural context.

1.2 Motivation of the Study

The study is motivated by the increasing prevalence of mental health issues among students and the lack of accessible, efficient digital support systems. It seeks to empower students by providing an online platform for mental health and performance counselling. By adopting a technology-driven approach, the study aims to reduce barriers to accessing professional support, enhance timely interventions, and promote academic and personal development.

1.3 Significance of the Study

The study is significant for several stakeholders:

- **Students:** Improved access to counselling services, increased academic performance, and better mental well-being.
- **Counsellors:** Streamlined management of appointments, easier tracking of student progress, and ability to provide timely interventions.

1.4 Scope of the Study

The scope of the study includes the design, development, and evaluation of a web-based mental health and performance counselling application for students at Cavendish University Zambia. The application will include modules for user registration, anonymous counselling requests, appointment scheduling, real-time chat with counsellors, and tracking academic performance metrics. The study is limited to the university environment and will not extend to external mental health institutions.

1.5 Problem Statement

Despite growing awareness of mental health challenges among university students, existing counselling services are insufficient to meet demand. Traditional counselling systems often suffer from limited accessibility, social stigma, and administrative inefficiencies. As a result, many students fail to receive timely support, which negatively impacts their academic performance, personal development, and overall well-being. A web-based platform tailored to student needs is necessary to improve access, confidentiality, and efficiency of counselling services.

1.6 Objectives

1.6.1 General Objective

To design, develop, and evaluate a student mental health and performance counselling web application that improves access to counselling services for students at Cavendish University Zambia.

1.6.2 Specific Objectives

1. To assess the mental health and counselling needs of students at Cavendish University Zambia.

2. To design and develop a user-friendly and secure web application for mental health and performance counselling.
3. To implement features for appointment scheduling, anonymous chats, and performance tracking.
4. To evaluate the effectiveness and usability of the developed application in enhancing student well-being and academic performance.
5. To integrate **self-help mental health resources** such as articles, videos, and wellness assessments.
6. To enable students to access support anytime, anywhere.

1.7 Research Questions

1. What are the primary mental health and counselling needs of students at Cavendish University Zambia?
2. How can a web-based application be designed and developed to provide accessible and effective counselling services?
3. What functionalities are most effective in enhancing counselling accessibility and academic performance tracking?
4. How effective is the developed system in improving student access to counselling and overall well-being?

1.8 Conceptual/Theoretical Framework

This study adopts the Health Belief Model (HBM) as its theoretical framework, which posits that individuals are more likely to engage in health-supportive behaviors when they perceive a risk and believe that a particular action will mitigate that risk. The Technology Acceptance Model (TAM) is also applied to assess students' willingness to adopt the web-based system. The conceptual framework illustrates the relationship between perceived mental health risks, perceived benefits of using the system, ease of use, and actual usage behavior.

[Insert Conceptual Framework Diagram Here]

1.9 Definition of Terms

- **Mental Health:** Psychological well-being enabling an individual to cope with normal stress and realize potential.
- **Counselling:** Professional support aimed at assisting individuals in coping with personal, social, or academic challenges.
- **Web Application:** A software application accessed via a web browser, designed to perform specific functions.
- **Performance Counselling:** Guidance provided to improve academic outcomes and personal growth.
- **User-Centered Design:** An iterative design process focusing on the needs, preferences, and limitations of end-users.
- **Iterative Development:** A methodology involving repeated cycles of prototyping, testing, and refinement.

1.10 Organization of the Report

The report is organized as follows:

- **Chapter One:** Introduces the problem, motivation, objectives, scope, and conceptual framework.
- **Chapter Two:** Reviews relevant literature on mental health counselling systems, identifies gaps, and discusses related works.
- **Chapter Three:** Describes the research methodology, data collection, system design, and development process.
- **References:** Lists all sources cited in the proposal following Harvard/APA style.

CHAPTER TWO: LITERATURE REVIEW

2.0 Overview

A literature review examines existing knowledge and research on a topic, identifying trends, gaps, and methodologies. It establishes the foundation for the proposed study, justifying the need for new research or system development.

2.1 Literature Review

Digital mental health solutions have been implemented globally to support student well-being. For example, online counselling platforms in the USA and UK have demonstrated improvements in student mental health outcomes (Smith et al., 2021; Johnson, 2020). These systems typically include appointment booking, chat support, and self-help resources. In African universities, adoption has been slower due to infrastructure and resource constraints, although pilot studies in Kenya and South Africa indicate positive impacts (Moyo, 2021; Karanja, 2020). In Zambia, literature on digital mental health interventions is scarce, highlighting a significant research gap.

2.2 Related Works

Several related works provide insights for this study:

1. **Student Counsel App (USA):** Provides chat-based counselling which improves access and reduces stigma.
2. **Mind Mate (UK):** Integrates mental health monitoring, guided exercises, and counsellor feedback.
3. **Kenya University Digital Counselling:** Pilot web platform with appointment booking and anonymous chat; positively influenced student stress management.
4. **South African University Wellbeing Platform:** Focuses on tracking student mental health metrics and providing personalized support.

These works inform the system design, highlighting features such as anonymity, usability, and data-driven decision-making.

2.3 Gaps in the Literature

While existing systems have improved mental health support elsewhere countrywide, few have been tailored to the Zambian higher education context. Gaps include lack of

localized content, limited integration with academic performance tracking, and insufficient studies on usability and effectiveness. This project aims to address these gaps by providing a culturally sensitive, accessible, and efficient web-based counselling solution for Cavendish University students.

CHAPTER THREE: METHODOLOGY AND SYSTEM DESIGN

3.0 Overview

This chapter outlines the methodology for conducting the study and the system design process for the web-based counselling application. It details the research design, data collection methods, and technical specifications.

3.1 Research Design

A mixed-method research design will be employed. Quantitative data will be gathered through surveys, while qualitative insights will be obtained via interviews. The system development will adopt an iterative, user-centered design approach, ensuring alignment with student needs and preferences.

3.2 Baseline Study

3.2.1 Data Collection

Data will be collected from students and counsellors using online questionnaires and semi-structured interviews. The survey will assess mental health challenges, preferences for counselling, and expectations for digital platforms. Interviews will explore experiences with existing counselling services and suggestions for improvement. Ethical considerations, such as informed consent and data confidentiality, will be strictly adhered to.

3.2.2 Research Approach

The study will adopt a user-centered, iterative approach. Prototypes of the web application will be developed, tested by students, and refined based on feedback, ensuring usability, accessibility, and effectiveness.

3.2.3 Development of the Application

The web application will be developed using modern web technologies including HTML5, CSS3, JavaScript, and backend frameworks such as PHP. A MySQL database will store user profiles, session data, and performance metrics. Security measures such as authentication, and data anonymization will be implemented.

3.2.4 System Design

The system will be modular, including the following components:

- **User Authentication Module:** Manages login, registration, and role-based access.
- **Counselling Module:** Allows students to request sessions, communicate with counsellors, and access resources.
- **Appointment Module:** Enables scheduling and notifications for sessions.
- **Performance Tracking Module:** Monitors academic metrics and correlates with counselling sessions.
- **Reporting Module:** Generates reports for counsellors and administrators.

3.3 System Design Diagrams

3.3.1 Context Diagram

Insert Context Diagram Here: Shows the web app interacting with students, counsellors, and administrators

3.3.2 System Software Level Architectural Design

Insert Architectural Diagram Here: Illustrates the frontend, backend, and database layers

3.3.3 Modular Design of the System Functions

Insert Modular Design Diagram Here: Depicts modules and their interactions

3.3.4 System Class Diagram

Insert Class Diagram Here: Shows classes like Student, Counsellor, Session, Appointment

3.3.5 System Data Model Design

Insert Data Model Diagram Here: Shows database tables, relationships, and attributes

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